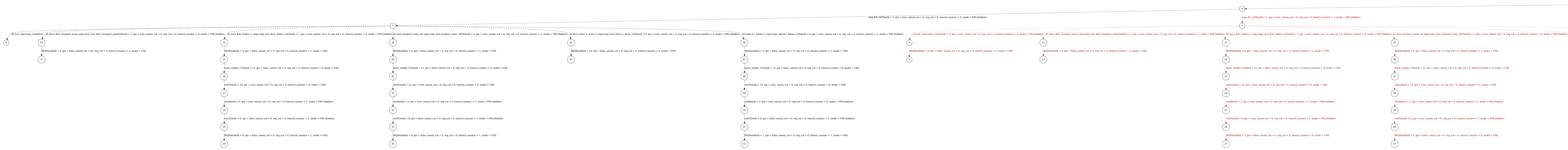
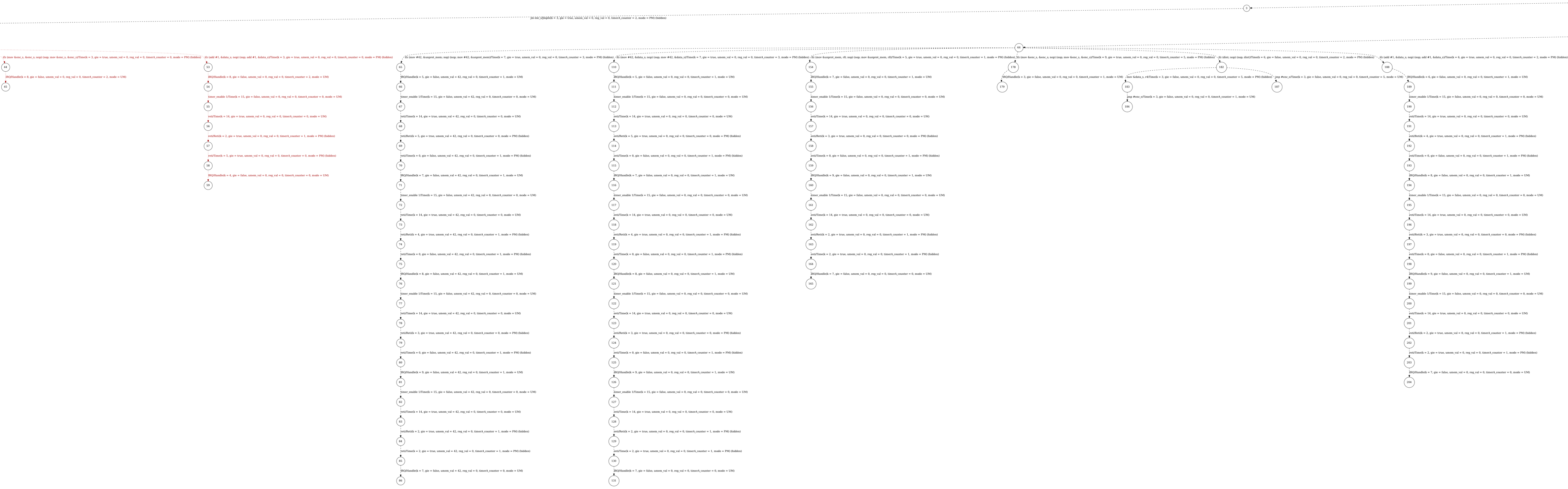
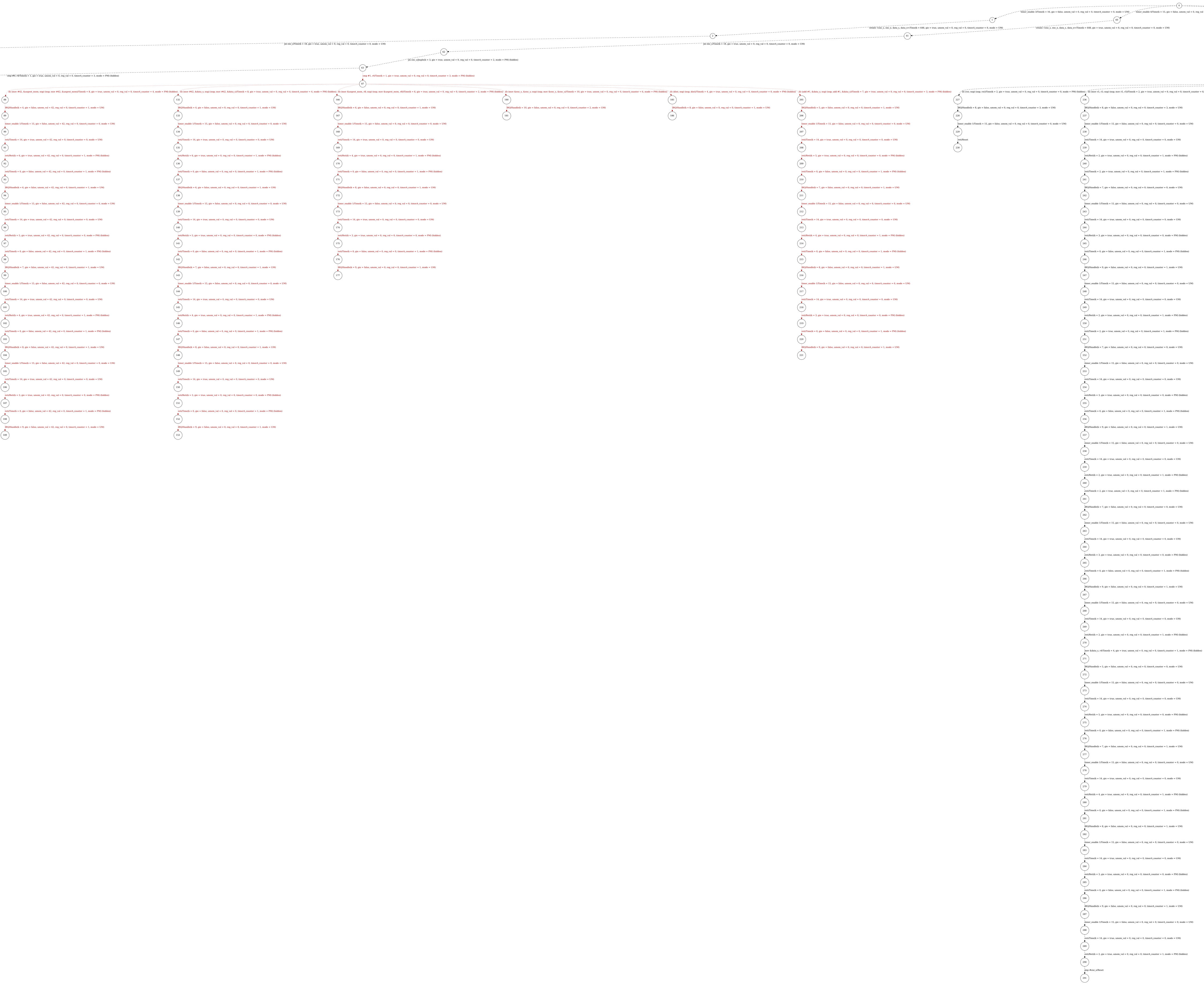
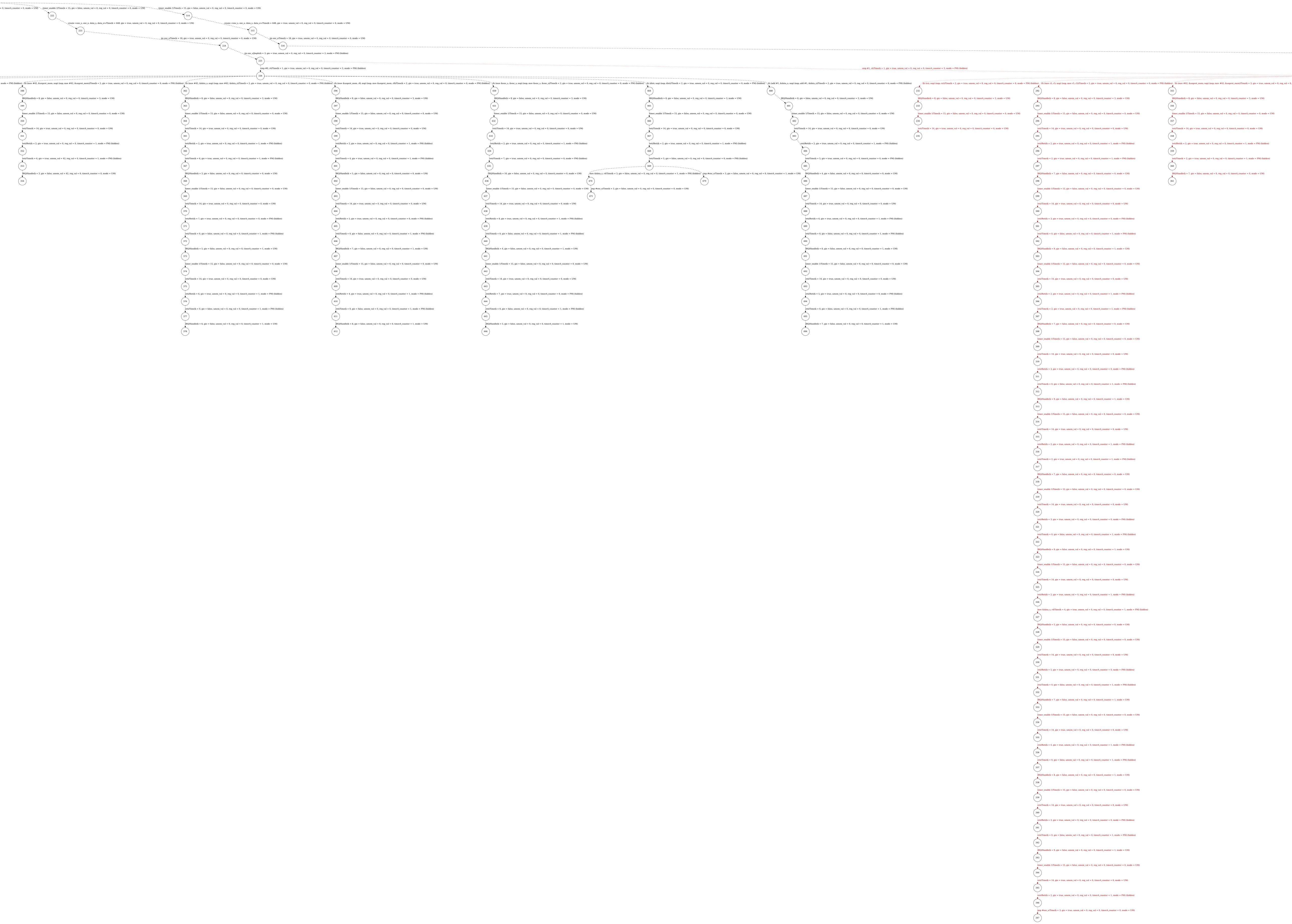
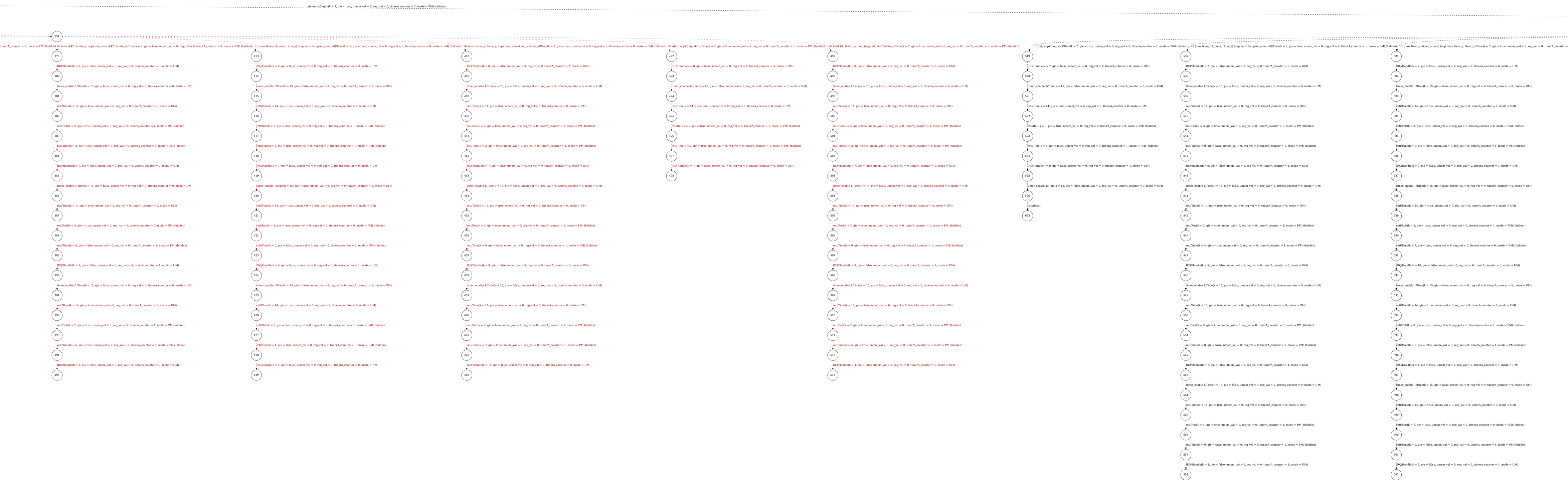
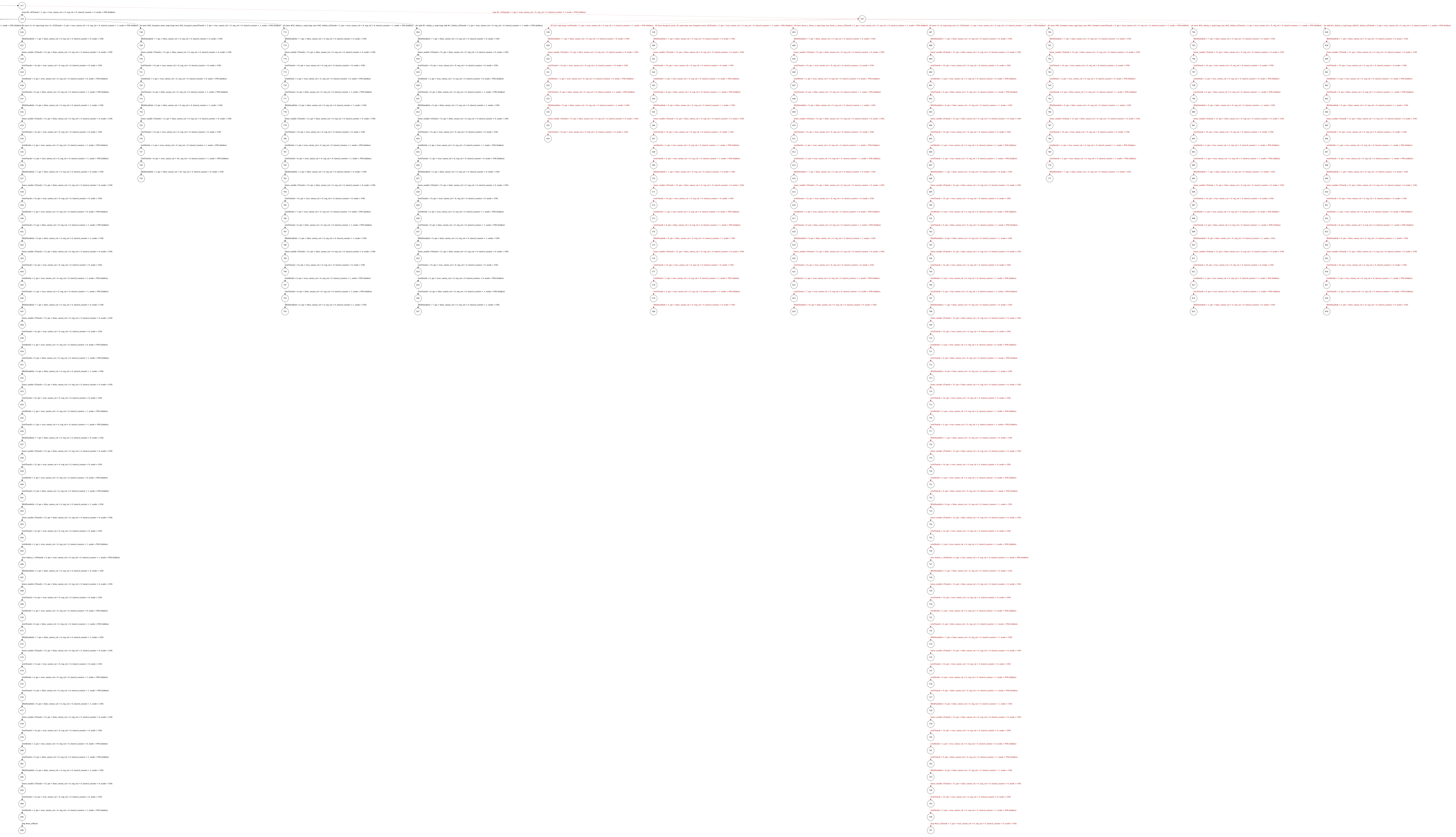

4
cmp #0, r4/Time(k = 1, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 3, mode = PM) (hidden)
cmp #1, r4/Time(k = 1, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 3, mode = PM) (hidden)
5
7
ifz (rst; nop) (nop; rst)/Reset
ifz (mov #42, &unprot_mem; nop) (nop; mov #42, &unprot_mem)/Time(k = 7, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 4, mode = PM) (hidden)
ifz (mov #42, &data_s; nop) (nop; mov #42, &data_s)/Time(k = 7, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 4, mode = PM) (hidden)
ifz (mov &unprot_mem, r8; nop) (nop; mov &unprot_mem, r8)/Time(k = 5, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 2, mode = PM) (hidden)
ifz (mov &enc_s, &enc_s; nop) (nop; mov &enc_s, &enc_s)/Time(k = 9, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
ifz (add #1, &data_s; nop) (nop; add #1, &data_s)/Time(k = 6, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 3, mode = PM) (hidden)
ifz (rst; nop) (nop; rst)/Time(k = 3, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
ifz (mov #42, &unprot_mem; nop) (nop; mov #42, &unprot_mem)/Time(k = 3, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
ifz (mov #42, &data_s; nop) (nop; mov #42, &data_s)/Time(k = 3, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
ifz (mov &unprot_mem, r8; nop) (nop; mov &unprot_mem, r8)/Time(k = 3, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
6
10
14
28
42
46
8
12
21
35
IRQ/Handle(k = 4, gie = false, umem_val = 42, reg_val = 0, timerA_counter = 2, mode = UM)
IRQ/Handle(k = 4, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 2, mode = UM)
IRQ/Handle(k = 6, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 2, mode = UM)
IRQ/Handle(k = 10, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 4, mode = UM)
IRQ/Handle(k = 5, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 2, mode = UM)
IRQ/Handle(k = 8, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 2, mode = UM)
IRQ/Handle(k = 8, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 2, mode = UM)
IRQ/Handle(k = 8, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 2, mode = UM)
IRQ/Handle(k = 8, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 2, mode = UM)
11
15
29
43
47
9
13
22
36
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
16
30
48
23
37
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
17
31
49
24
38
reti/Reti(k = 6, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
reti/Reti(k = 4, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
reti/Reti(k = 5, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
reti/Reti(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
reti/Reti(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
18
32
50
25
39
reti/Time(k = 0, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
reti/Time(k = 0, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
reti/Time(k = 0, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
reti/Time(k = 6, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
reti/Time(k = 4, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
19
33
51
26
40
IRQ/Handle(k = 6, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
IRQ/Handle(k = 8, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
IRQ/Handle(k = 7, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
IRQ/Handle(k = 3, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
IRQ/Handle(k = 5, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
20
34
52
27
41


3
jin enc_s/JmpIn(k = 3, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 2, mode = PM) (hidden)
64
ifz (mov &enc_s, &enc_s; nop) (nop; mov &enc_s, &enc_s)/Time(k = 3, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
44
IRQ/Handle(k = 8, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 2, mode = UM)
45
ifz (add #1, &data_s; nop) (nop; add #1, &data_s)/Time(k = 3, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
53
IRQ/Handle(k = 8, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 2, mode = UM)
54
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
55
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
56
reti/Reti(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
57
reti/Time(k = 5, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
58
IRQ/Handle(k = 4, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
59
ifz (mov #42, &unprot_mem; nop) (nop; mov #42, &unprot_mem)/Time(k = 7, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 3, mode = PM) (hidden)
65
IRQ/Handle(k = 5, gie = false, umem_val = 42, reg_val = 0, timerA_counter = 1, mode = UM)
66
timer_enable 1/Time(k = 15, gie = false, umem_val = 42, reg_val = 0, timerA_counter = 0, mode = UM)
67
reti/Time(k = 14, gie = true, umem_val = 42, reg_val = 0, timerA_counter = 0, mode = UM)
68
reti/Reti(k = 5, gie = true, umem_val = 42, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
69
reti/Time(k = 0, gie = false, umem_val = 42, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
70
IRQ/Handle(k = 7, gie = false, umem_val = 42, reg_val = 0, timerA_counter = 1, mode = UM)
71
timer_enable 1/Time(k = 15, gie = false, umem_val = 42, reg_val = 0, timerA_counter = 0, mode = UM)
72
reti/Time(k = 14, gie = true, umem_val = 42, reg_val = 0, timerA_counter = 0, mode = UM)
73
reti/Reti(k = 4, gie = true, umem_val = 42, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
74
reti/Time(k = 0, gie = false, umem_val = 42, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
75
IRQ/Handle(k = 8, gie = false, umem_val = 42, reg_val = 0, timerA_counter = 1, mode = UM)
76
timer_enable 1/Time(k = 15, gie = false, umem_val = 42, reg_val = 0, timerA_counter = 0, mode = UM)
77
reti/Time(k = 14, gie = true, umem_val = 42, reg_val = 0, timerA_counter = 0, mode = UM)
78
reti/Reti(k = 3, gie = true, umem_val = 42, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
79
reti/Time(k = 0, gie = false, umem_val = 42, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
80
IRQ/Handle(k = 9, gie = false, umem_val = 42, reg_val = 0, timerA_counter = 1, mode = UM)
81
timer_enable 1/Time(k = 15, gie = false, umem_val = 42, reg_val = 0, timerA_counter = 0, mode = UM)
82
reti/Time(k = 14, gie = true, umem_val = 42, reg_val = 0, timerA_counter = 0, mode = UM)
83
reti/Reti(k = 2, gie = true, umem_val = 42, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
84
reti/Time(k = 2, gie = true, umem_val = 42, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
85
IRQ/Handle(k = 7, gie = false, umem_val = 42, reg_val = 0, timerA_counter = 0, mode = UM)
86
ifz (mov #42, &data_s; nop) (nop; mov #42, &data_s)/Time(k = 7, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 3, mode = PM) (hidden)
110
IRQ/Handle(k = 5, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
111
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
112
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
113
reti/Reti(k = 5, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
114
reti/Time(k = 0, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
115
IRQ/Handle(k = 7, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
116
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
117
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
118
reti/Reti(k = 4, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
119
reti/Time(k = 0, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
120
IRQ/Handle(k = 8, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
121
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
122
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
123
reti/Reti(k = 3, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
124
reti/Time(k = 0, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
125
IRQ/Handle(k = 9, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
126
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
127
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
128
reti/Reti(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
129
reti/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
130
IRQ/Handle(k = 7, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
131
ifz (mov &unprot_mem, r8; nop) (nop; mov &unprot_mem, r8)/Time(k = 5, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
154
IRQ/Handle(k = 7, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
155
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
156
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
157
reti/Reti(k = 3, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
158
reti/Time(k = 0, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
159
IRQ/Handle(k = 9, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
160
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
161
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
162
reti/Reti(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
163
reti/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
164
IRQ/Handle(k = 7, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
165
ifz (mov &enc_s, &enc_s; nop) (nop; mov &enc_s, &enc_s)/Time(k = 9, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 5, mode = PM) (hidden)
178
IRQ/Handle(k = 3, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
179
ifz (dint; nop) (nop; dint)/Time(k = 6, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 2, mode = PM) (hidden)
182
mov &data_s, r4/Time(k = 3, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 5, mode = PM) (hidden)
183
jmp #enc_e/Time(k = 3, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
184
jmp #enc_e/Time(k = 3, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 5, mode = UM)
187
ifz (add #1, &data_s; nop) (nop; add #1, &data_s)/Time(k = 6, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 2, mode = PM) (hidden)
188
IRQ/Handle(k = 6, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
189
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
190
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
191
reti/Reti(k = 4, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
192
reti/Time(k = 0, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
193
IRQ/Handle(k = 8, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
194
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
195
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
196
reti/Reti(k = 3, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
197
reti/Time(k = 0, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
198
IRQ/Handle(k = 9, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
199
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
200
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
201
reti/Reti(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
202
reti/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
203
IRQ/Handle(k = 7, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
204


0
timer_enable 3/Time(k = 16, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
timer_enable 4/Time(k = 15, gie = false, umem_val = 0, reg_val
1
60
create <enc_s, enc_e, data_s, data_e>/Time(k = 648, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
create <enc_s, enc_e, data_s, data_e>/Time(k = 648, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
2
61
jin enc_s/Time(k = 18, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
jin enc_s/Time(k = 18, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
62
jin enc_s/JmpIn(k = 3, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 2, mode = PM) (hidden)
63
cmp #0, r4/Time(k = 1, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 3, mode = PM) (hidden)
cmp #1, r4/Time(k = 1, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 3, mode = PM) (hidden)
87
ifz (mov #42, &unprot_mem; nop) (nop; mov #42, &unprot_mem)/Time(k = 8, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 4, mode = PM) (hidden)
ifz (mov #42, &data_s; nop) (nop; mov #42, &data_s)/Time(k = 8, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 4, mode = PM) (hidden)
ifz (mov &unprot_mem, r8; nop) (nop; mov &unprot_mem, r8)/Time(k = 6, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 2, mode = PM) (hidden)
ifz (mov &enc_s, &enc_s; nop) (nop; mov &enc_s, &enc_s)/Time(k = 10, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 6, mode = PM) (hidden)
ifz (dint; nop) (nop; dint)/Time(k = 4, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
ifz (add #1, &data_s; nop) (nop; add #1, &data_s)/Time(k = 7, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 3, mode = PM) (hidden)
ifz (rst; nop) (nop; rst)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
ifz (mov r5, r5; nop) (nop; mov r5, r5)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0,
88
IRQ/Handle(k = 4, gie = false, umem_val = 42, reg_val = 0, timerA_counter = 1, mode = UM)
89
timer_enable 1/Time(k = 15, gie = false, umem_val = 42, reg_val = 0, timerA_counter = 0, mode = UM)
90
reti/Time(k = 14, gie = true, umem_val = 42, reg_val = 0, timerA_counter = 0, mode = UM)
91
reti/Reti(k = 6, gie = true, umem_val = 42, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
92
132
IRQ/Handle(k = 4, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
133
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
134
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
135
reti/Reti(k = 6, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
136
166
IRQ/Handle(k = 6, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
167
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
168
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
169
reti/Reti(k = 4, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
170
180
IRQ/Handle(k = 10, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 2, mode = UM)
181
185
IRQ/Handle(k = 8, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
186
205
IRQ/Handle(k = 5, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
206
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
207
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
208
reti/Reti(k = 5, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
209
227
IRQ/Handle(k = 8, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 3, mode = UM)
228
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
229
reti/Reset
230
236
IRQ/Handle(k = 8, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 3, mode = UM)
237
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
238
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
239
reti/Reti(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
240


= 0, timerA_counter = 0, mode = UM)
timer_enable 2/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
create <enc_s, enc_e, data_s, data_e>/Time(k = 648, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
create <enc_s, enc_e, data_s, data_e>/Time(k = 648, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
jin enc_s/Time(k = 18, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
jin enc_s/Time(k = 18, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
jin enc_s/JmpIn(k = 3, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 2, mode = PM) (hidden)
cmp #0, r4/Time(k = 1, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 3, mode = PM) (hidden)
cmp #1, r4/Time(k = 1, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 3, mode = PM) (hidden)
mode = PM) (hidden)
ifz (mov #42, &unprot_mem; nop) (nop; mov #42, &unprot_mem)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
ifz (mov #42, &data_s; nop) (nop; mov #42, &data_s)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
ifz (mov &unprot_mem, r8; nop) (nop; mov &unprot_mem, r8)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
ifz (mov &enc_s, &enc_s; nop) (nop; mov &enc_s, &enc_s)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
ifz (dint; nop) (nop; dint)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
ifz (add #1, &data_s; nop) (nop; add #1, &data_s)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
ifz (rst; nop) (nop; rst)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
ifz (mov r5, r5; nop) (nop; mov r5, r5)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
ifz (mov #42, &unprot_mem; nop) (nop; mov #42, &unprot_mem)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0,
IRQ/Handle(k = 8, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 3, mode = UM)
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)


jin enc_s/JmpIn(k = 3, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 2, mode = PM) (hidden)
231
timerA_counter = 0, mode = PM) (hidden)
ifz (mov #42, &data_s; nop) (nop; mov #42, &data_s)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
379
IRQ/Handle(k = 8, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 3, mode = UM)
380
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
381
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
382
reti/Reti(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
383
reti/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
384
IRQ/Handle(k = 7, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
385
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
386
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
387
reti/Reti(k = 3, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
388
reti/Time(k = 0, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
389
IRQ/Handle(k = 9, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
390
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
391
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
392
reti/Reti(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
393
reti/Time(k = 6, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
394
IRQ/Handle(k = 3, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
395
ifz (mov &unprot_mem, r8; nop) (nop; mov &unprot_mem, r8)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
413
IRQ/Handle(k = 8, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 3, mode = UM)
414
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
415
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
416
reti/Reti(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
417
reti/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
418
IRQ/Handle(k = 7, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
419
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
420
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
421
reti/Reti(k = 3, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
422
reti/Time(k = 0, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
423
IRQ/Handle(k = 9, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
424
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
425
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
426
reti/Reti(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
427
reti/Time(k = 4, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
428
IRQ/Handle(k = 5, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
429
ifz (mov &enc_s, &enc_s; nop) (nop; mov &enc_s, &enc_s)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
447
IRQ/Handle(k = 8, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 3, mode = UM)
448
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
449
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
450
reti/Reti(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
451
reti/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
452
IRQ/Handle(k = 7, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
453
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
454
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
455
reti/Reti(k = 3, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
456
reti/Time(k = 0, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
457
IRQ/Handle(k = 9, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
458
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
459
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
460
reti/Reti(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
461
reti/Time(k = 7, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
462
IRQ/Handle(k = 10, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
463
ifz (dint; nop) (nop; dint)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
472
IRQ/Handle(k = 8, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 3, mode = UM)
473
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
474
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
475
reti/Reti(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
476
reti/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
477
IRQ/Handle(k = 7, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
478
ifz (add #1, &data_s; nop) (nop; add #1, &data_s)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
497
IRQ/Handle(k = 8, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 3, mode = UM)
498
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
499
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
500
reti/Reti(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
501
reti/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
502
IRQ/Handle(k = 7, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
503
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
504
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
505
reti/Reti(k = 3, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
506
reti/Time(k = 0, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
507
IRQ/Handle(k = 9, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
508
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
509
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
510
reti/Reti(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
511
reti/Time(k = 5, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
512
IRQ/Handle(k = 4, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
513
ifz (rst; nop) (nop; rst)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
519
IRQ/Handle(k = 7, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
520
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
521
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
522
reti/Reti(k = 3, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
523
reti/Time(k = 0, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
524
IRQ/Handle(k = 9, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
525
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
526
reti/Reset
625
ifz (mov &unprot_mem, r8; nop) (nop; mov &unprot_mem, r8)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
537
IRQ/Handle(k = 7, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
538
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
539
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
540
reti/Reti(k = 3, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
541
reti/Time(k = 0, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
542
IRQ/Handle(k = 9, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
543
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
544
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
545
reti/Reti(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
546
reti/Time(k = 4, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
547
IRQ/Handle(k = 5, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
548
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
549
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
550
reti/Reti(k = 5, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
551
reti/Time(k = 0, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
552
IRQ/Handle(k = 7, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
553
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
554
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
555
reti/Reti(k = 4, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
556
reti/Time(k = 0, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
557
IRQ/Handle(k = 8, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
558
ifz (mov &enc_s, &enc_s; nop) (nop; mov &enc_s, &enc_s)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter =
581
IRQ/Handle(k = 7, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
582
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
583
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
584
reti/Reti(k = 3, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
585
reti/Time(k = 0, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
586
IRQ/Handle(k = 9, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
587
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
588
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
589
reti/Reti(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
590
reti/Time(k = 7, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
591
IRQ/Handle(k = 10, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
592
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
593
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
594
reti/Reti(k = 8, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
595
reti/Time(k = 0, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
596
IRQ/Handle(k = 4, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
597
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
598
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
599
reti/Reti(k = 7, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
600
reti/Time(k = 0, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
601
IRQ/Handle(k = 5, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)
602


517
cmp #0, r4/Time(k = 1, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 3, mode = PM) (hidden)
518
ifz (mov r5, r5; nop) (nop; mov r5, r5)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
ifz (mov #42, &unprot_mem; nop) (nop; mov #42, &unprot_mem)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
ifz (mov #42, &data_s; nop) (nop; mov #42, &data_s)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
ifz (add #1, &data_s; nop) (nop; add #1, &data_s)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
cmp #1, r4/Time(k = 1, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 3, mode = PM) (hidden)
527
ifz (rst; nop) (nop; rst)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
ifz (mov &unprot_mem, r8; nop) (nop; mov &unprot_mem, r8)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
ifz (mov &enc_s, &enc_s; nop) (nop; mov &enc_s, &enc_s)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
ifz (mov r5, r5; nop) (nop; mov r5, r5)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
ifz (mov #42, &unprot_mem; nop) (nop; mov #42, &unprot_mem)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
ifz (mov #42, &data_s; nop) (nop; mov #42, &data_s)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
ifz (add #1, &data_s; nop) (nop; add #1, &data_s)/Time(k = 2, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
IRQ/Handle(k = 7, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
timer_enable 1/Time(k = 15, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
reti/Time(k = 14, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = UM)
reti/Reti(k = 3, gie = true, umem_val = 0, reg_val = 0, timerA_counter = 0, mode = PM) (hidden)
reti/Time(k = 0, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = PM) (hidden)
IRQ/Handle(k = 9, gie = false, umem_val = 0, reg_val = 0, timerA_counter = 1, mode = UM)